Lecture Scribe: Transformation of Random Variables

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Lecture Scribe: CSE400 - Fundamentals of Probability in Computing

Lecture 12: Transformation of Random Variables
Instructor: Dhaval Patel, PhD
Date: February 12, 2026

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1. Introduction and Objectives

The primary goal of this lecture is the **learning of transformation techniques for random variables**. This involves understanding how to find the probability distribution of a new random variable (Y) that is defined as a function of an existing random variable (X).

Key Concepts:

- **A Priori Knowledge:** We assume the Probability Density Function (PDF), denoted as $f_X(x)$, and the Cumulative Distribution Function (CDF), $F_X(x)$, of the original random variable X are already known.
- **The Transformation:** A new random variable is defined by the function $y = g(x)$.
- **Derived Distributions:** The objective is to determine the new PDF, $f_Y(y)$, or CDF, $F_Y(y)$.

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2. Transformation of a Single Random Variable

To find the PDF of Y when $y = g(x)$, we follow a systematic derivation based on the CDF.

Step 1: Find the CDF of Y

The CDF of Y is defined as the probability that Y is less than or equal to a specific value y:

$$F_Y(y) = Pr(Y \leq y) = Pr(g(x) \leq y)$$

If the function $g(x)$ is monotonically increasing, we can use the inverse function g^{-1}:

$$F_Y(y) = Pr(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

Step 2: Differentiate to find the PDF

To find the PDF $f_Y(y)$, we differentiate the CDF with respect to y:

$$f_Y(y) = \frac{d}{dy}[F_X(g^{-1}(y))]$$

Using the chain rule, this results in:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Alternatively, expressed in terms of x:

$$f_Y(y) = f_X(x) \left| [cite_s tart] \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

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3. Illustrative Example: Uniform to Sine Transformation

Problem: Given X is a uniformly distributed random variable on the interval $(-1, 1)$, find the PDF of $Y = \sin(\frac{\pi x}{2})$.

Solution:

1. **Known PDF of X:**

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

2. **Inverse Transformation:**
   Given $y = \sin(\frac{\pi x}{2})$, then $x = \frac{2}{\pi} \sin^{-1}(y)$.
3. **Derivative Calculation:**

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{1 - y^2}$$

4. **Final PDF of Y:**
   Applying the formula $f_Y(y) = f_X(x) \cdot |\frac{dx}{dy}|$:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{1 - y^2} = \frac{1}{\pi \quad 1 - y^2}$$

   Limits: Since $x \in (-1, 1)$, y ranges from $\sin(-\pi/2)$ to $\sin(\pi/2)$, which is $-1 < y < 1$.

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4. Function of Two Random Variables

This section deals with joint transformations where a new random variable Z is a function of two variables: $Z = g(X, Y)$.

Case Study: $Z = X + Y$

To find the PDF $f_Z(z)$:

1. **Find the CDF:**

$$F_Z(z) = Pr(X + Y \leq z)$$

2. **Set up the Double Integral:**

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

3. **Differentiate using Leibniz's Rule:**
   Leibniz's rule for differentiating an integral is defined as:

$$\frac{d}{dx}\left[\int_{a(x)}^{b(x)} h(x, y)\, dy\right] = \frac{d}{dx}b(x) \cdot h(x, b(x)) - \frac{d}{dx}a(x) \cdot h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x}h(x, y)\, dy$$

4. **Resulting PDF (Convolution):**
   If X and Y are independent, $f_{XY}(x, y) = f_X(x)f_Y(y)$. The PDF of their sum is the convolution integral:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)\, dx$$

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5. Further Derived Distributions

The lecture notes briefly outline other transformations of two variables:

- **Difference ($Z = X - Y$):** The procedure is similar to the sum, starting with $F_Z(z) = Pr(X - Y \leq z)$.
- **Ratio ($Z = X/Y$):** Calculated via $Pr(X/Y \leq z)$.
- **Square Root of Sum of Squares:** Finding the PDF of $R = \quad X^2 + Y^2$.

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6. Advanced Examples

- **Sum of Normal Variables:** If $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ are independent, then $Z = X + Y$ follows $N(0, 2)$. This is proven by substituting the Gaussian PDF into the convolution integral.
- **Sum of Exponential Variables:** If X and Y are independent exponential random variables with parameter λ, the PDF of $Z = X + Y$ is found using the integral of $\lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)}$ over the interval $[0, z]$.

Next Step: Would you like me to generate practice problems based on the transformations discussed in these slides, such as the sum of exponential variables?

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Lecture Scribe: CSE400 - Fundamentals of Probability in Computing

Lecture 12: Transformation of Random Variables Instructor: Dhaval Patel, PhD **Date:** February 12, 2026

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1. Introduction and Objectives

The primary objective of this lecture is the **learning of transformation techniques for random variables**. This involves determining the probability distribution of a new random variable (Y) that is defined as a function of an existing random variable (X).

Key Assumptions and A Priori Knowledge

- The **Probability Density Function (PDF)**, $f_X(x)$, and the **Cumulative Distribution Function (CDF)**, $F_X(x)$, of the original random variable X are assumed to be known beforehand (a priori).
- The new random variable is defined by the transformation $y = g(x)$.
- The goal is to find the new PDF $f_Y(y)$ or CDF $F_Y(y)$.

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2. Transformation of a Single Random Variable

To derive the PDF of Y, we typically follow a two-step process starting from the CDF.

Step 1: Find the CDF of Y ($F_Y(y)$)

The CDF of Y is defined as the probability that the random variable Y is less than or equal to a specific value y:

$$F_Y(y) = Pr(Y \leq y) = Pr(g(x) \leq y)$$

If the function $g(x)$ is monotonically increasing, we can use the inverse function $g^{-1}(y)$:

$$F_Y(y) = Pr(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

If the function is decreasing, the inequality flips:

$$F_Y(y) = Pr(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Step 2: Differentiate to find the PDF ($f_Y(y)$)

To find the PDF, we differentiate the CDF with respect to y:

$$f_Y(y) = \frac{d}{dy}[F_X(g^{-1}(y))]$$

Using the chain rule, this becomes:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

In terms of the derivative of x with respect to y, this is expressed as:

$$f_Y(y) = f_X(x) \left| [cite_s tart] \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

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3. Illustrative Example: Uniform to Sine Transformation

Problem: Given X is a uniformly distributed random variable on the interval $(-1, 1)$, find the PDF of $Y = \sin(\frac{\pi x}{2})$.

Solution:

1. **Known PDF of X:**

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

2. **Inverse Transformation:**
   Given $y = \sin(\frac{\pi x}{2})$, the inverse is $x = \frac{2}{\pi} \sin^{-1}(y)$.
3. **Find the Derivative:**

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{1 - y^2}$$

4. **Calculate $f_Y(y)$:**

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right| [cite_s tart] = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{1 - y^2} = \frac{1}{\pi} \frac{1}{1 - y^2}$$

5. **Limits:**
   Since $x \in (-1, 1)$, then $y \in (-1, 1)$ because $\sin(-\pi/2) = -1$ and $\sin(\pi/2) = 1$.

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4. Function of Two Random Variables

This involves joint transformations where a new random variable Z is a function of two variables: $Z = g(X, Y)$.

Case: $Z = X + Y$

To find the PDF $f_Z(z)$:

1. **Find the CDF:**

$$F_Z(z) = Pr(X + Y \leq z)$$

2. **Double Integral Representation:**

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy$$

3. **Differentiate using Leibniz's Rule:**
   Leibniz's rule states:

$$\frac{d}{dx}\left[\int_{a(x)}^{b(x)} h(x, y)\, dy\right] = \frac{d}{dx}b(x) \cdot h(x, b(x)) - \frac{d}{dx}a(x) \cdot h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x}h(x, y)\, dy$$

4. **Convolution Integral:**
   If X and Y are independent, the PDF of their sum is the convolution of their individual PDFs:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$$

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5. Other Transformations of Two Variables

The same logic applies to other functions:

- **Difference ($Z = X - Y$):** $F_Z(z) = Pr(X - Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y)\, dx\, dy$.
- **Ratio ($Z = X/Y$):** $F_Z(z) = Pr(X/Y \leq z)$.
- **Square Root of Sum of Squares:** $R = \quad X^2 + Y^2$.

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6. Advanced Examples from the PDF

- **Sum of Normal Variables:** If $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ are independent, then $Z = X + Y \sim N(0, 2)$. This is proven by integrating the product of two Gaussian PDFs.
- **Sum of Exponential Variables:** If X and Y are independent exponential random variables with parameter λ, the PDF $f_Z(z)$ is found using the convolution integral of $\lambda e^{-\lambda x}$ and $\lambda e^{-\lambda(z-x)}$.

Next Step: Would you like me to walk through the step-by-step differentiation of the sum $Z = X + Y$ using Leibniz's rule as shown on page 9?

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Lecture Scribe: CSE400 - Fundamentals of Probability in Computing

Lecture 11 & 12: Transformation of Random Variables Instructor: Dhaval Patel, PhD **Date:** February 10, 2026

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1. Introduction and Objectives

The primary objective of this lecture is the **learning of transformation techniques for random variables**. This involves determining the probability distribution of a new random variable (Y) derived from an existing one (X).

Core Assumptions

- The **Probability Density Function (PDF)**, $f_X(x)$, of the original random variable X is known a priori.
- The **Cumulative Distribution Function (CDF)**, $F_X(x)$, of X is also known.
- A new random variable Y is defined by a functional relationship: $y = g(x)$.

Objectives

- Determine how to find the new distribution $f_Y(y)$ or $F_Y(y)$.
- Understand joint transformations for two random variables.

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2. Transformation of a Single Random Variable

The derivation of the new PDF follows a systematic two-step process.

Step 1: Find the Cumulative Distribution Function (CDF)

The CDF of the new variable Y, $F_Y(y)$, is the probability that Y is less than or equal to a value y:

$$F_Y(y) = Pr(Y \leq y) = Pr(g(x) \leq y) \text{ [cite: 60]}$$

If the function $g(x)$ is monotonically increasing:

$$F_Y(y) = Pr(x \leq g^{-1}(y)) = F_X(g^{-1}(y)) \text{ [cite: 61]}$$

If the function $g(x)$ is decreasing:

$$F_Y(y) = Pr(x \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)) \text{ [cite: 70]}$$

Step 2: Differentiate to find the PDF

To find the PDF $f_Y(y)$, differentiate the CDF with respect to y:

$$f_Y(y) = \frac{d}{dy}[F_X(g^{-1}(y))] \text{ [cite: 52]}$$

Using the chain rule, this results in:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) \text{ [cite: 54]}$$

Or more generally, accounting for both increasing and decreasing functions:

$$f_Y(y) = f_X(x) \left| [cite_s tart] \frac{dx}{dy} \right|_{x=g^{-1}(y)} \text{ [cite: 71, 90]}$$

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3. Illustrative Example: Uniform to Sine Transformation

Problem: Given X is a uniformly distributed random variable on the interval $(-1, 1)$, find the PDF of $Y = \sin(\frac{\pi x}{2})$.

Solution:

1. **Original PDF of X:** $f_X(x) = \frac{1}{2}$ for $-1 < x < 1$, and 0 otherwise.
2. **Inverse Function:** $x = \frac{2}{\pi} \sin^{-1}(y)$.
3. **Derivative:** $\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{1-y^2}$.
4. **Resulting PDF:**

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{1-y^2} = \frac{1}{\pi \quad 1-y^2} \text{ [cite: 92]}$$

5. **Limits:** For $x = -1$, $y = \sin(-\pi/2) = -1$. For $x = 1$, $y = 1$. Thus, $-1 < y < 1$.

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4. Function of Two Random Variables

This section addresses joint transformations where $Z = g(X, Y)$.

Detailed Derivation: Sum of Two RVs ($Z = X + Y$)

To find the PDF $f_Z(z)$ for the sum $Z = X + Y$:

1. **Determine the CDF:**

$$F_Z(z) = Pr(X + Y \leq z) \text{ [cite: 121]}$$

2. **Set up the Double Integral:**

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y)\, dx\, dy \text{ [cite: 123]}$$

Application of Leibniz's Rule

To differentiate the integral, we use Leibniz's rule:

$$\frac{d}{dx} \left[\int_{a(x)}^{b(x)} h(x, y)\, dy \right] = \frac{d}{dx} b(x) \cdot h(x, b(x)) - \frac{d}{dx} a(x) \cdot h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)\, dy \text{ [cite: 133]}$$

Applying this to $F_Z(z)$ yields the **Convolution Integral**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y)\, dy \text{ [cite: 143]}$$

If X and Y are independent:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx \text{ [cite: 154]}$$

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5. Advanced Applications

Sum of Independent Normal Random Variables

- **Assumptions:** $X \sim N(0,1)$ and $Y \sim N(0,1)$ are independent.
- **Proof:** By substituting the Gaussian PDF $f_X(x) = \frac{1}{2\pi} e^{-x^2/2}$ into the convolution integral , it can be shown that $Z = X + Y \sim N(0,2)$.
- **Result:** The sum follows a normal distribution with variance $\sigma^2 = 2$.

Sum of Independent Exponential Random Variables

- **Assumptions:** X and Y are exponential random variables with parameter λ.
- **Method:** Use the convolution integral with $f_X(x) = \lambda e^{-\lambda x}$ and $f_Y(y) = \lambda e^{-\lambda y}$.
- **Integral:** $f_Z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)}\, dx$.

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6. Summary of Other Transformations

- **Difference ($Z = X - Y$):** $F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x,y)\, dx\, dy$.
- **Ratio ($Z = X/Y$):** $F_Z(z) = Pr(X/Y \leq z)$.
- **Square Root of Sum of Squares:** $R = \quad X^2 + Y^2$.

Next Step: Would you like me to create a summary table comparing the convolution integrals for the sum ($X + Y$) and difference ($X - Y$) based on these derivations?